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MIL-W-8939A(MI)
20 July 1990
SUPERSEDING
MIL-W-8939 (ASG)
11 January 1965

# MILITARY SPECIFICATION

## WELDING, RESISTANCE, ELECTRONIC CIRCUIT MODULES

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available fox use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification. establishes minimum requirements for resistance welding of electronic circuit module interconnections and component leads . It covers certification and qualification of welding machines and defines the requirements for quality control of resistance-welded circuits.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Spacifications, standard, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein Unless ohterwise specified the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-1276 - Leads for Electronic Component Parts

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3439

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(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications . The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

BULL 34 - Standard Resistance Welding Electrode Numbers and Alloys, Manufacturer's Cross Reference of

(Application for copies should be addressed to the Resistance Welders Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

[illegible] Method of Preparation of Metallographic Specimens

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedenced. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Applicability. The requirement specified herein are considered applicable in most of the conditions encountered in resistance welding of electronic circuits. When special conditions are encountered which render the requirements of this specification inapplicable, the contractor shall Submit, at the direction of the contracting officer, alternate procedures and requirements for approval by the procuring activity.

3.2 Materials. This specification covers resistance welding of the following materials for electronic use:

Group A - Pure nickel ribbon and wire.
Group B - Nickel-base and iron-base alloys.
Group C - Copper and copper alloys.

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3.2.1 Electronic component lead materials. Electronic components to be resistance welded shall contain lead materials conforming to MIL-STD-1276 wherever possible. Other types of leads shall be used only if they have been found suitable for the purpose, and if weld schedules can be developed which conform to applicable quality assurance provisions of this specification.

3.3 Fabrication and procedure. Welded electronic circuit modules and assemblies shall be fabricated in accordance with the applicable drawings or documents.

3.4 Equipment.

3.4.1 Welding machine. The welding machine shall consist of a suitable source of electrical energy, suitable electrodes, and a means of reliably controlling and indicating the magnitude of the current and welding force. The force and current controls shall operate so that no current flows until the welding force is applied at the welding electrodes. The current shall be cut off before the force is removed.

3.4.2 Electrodes. Suitable electrode material, identified by Resistance Welder Manufacturers Association (RWMA) class (see RWMA Bull 34), shall be used to perform welding in conformance with this specification. Electrodes with insulated shanks shall be used where required.

3.4.3 Shear testing machines. Weld shear testing machines shall be provided as required. All shear testing machines snail be accurate within ±2 percent of the indicated reading. Shear test machines shall be checked for accuracy at intervals not greater than three months, whenever any significant operational changes are made thereto, or when relocated.

3.5 Jigs and fixtures. All tooling that is required to position welds or assist in the assembly of welded parts that must pass through the magnetic field during the welding operation in a manner that would cause magnetic disturbances to the welding cycle shall be made of nonmagnetic materials insofar as practical. Jigs and fixtures shall be so designed that no welding current shunts through the locating devices rather than the workplaces.

3.6 Maintenance of equipment. Unless otherwise specified, each item of equipment shall be inspected periodically as recommended by the manufacturer. Adequate maintenance shall be furnished, and defective parts affecting machine operation shall be replaced before production welding is resumed.

3.7 Weld standards.

3.7.1 External appearance of welds.

3.7.1.1 Defects. The surface of all welds shall be free from cracks and voids when examined at a minimum magnification of 10 diameters. The contractor shall establish limits for other external appearance criteria, such as excessive expulsion, bulging, and the like. Such limits for electrode indentation, embedment, and set-down shall be not greater than either 50 percent of the thickness of the member most. affected or 35 percent of the total thickness of both materials.

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3.7.2 Metallographic. When examined at a minimum magnification of 50 diameters, welds shall be free from cracks. No weld shall contain voids which, considered together, exceed in total length 10 percent of the interface length. No void shall be closer than 10 percent of the interface length to the end of the interface. Electrode indentation, embedment, and set-down shall not exceed that allowed by 3.7.1.1.

3.7.3 Machine qualification test specimen.

3.7.3.1 Weld strength. The strength of each weld shall be equal to or greater than 80 percent of the minimum parent metal breaking strength.

3.7.3.2 Coefficient. The coefficient of variation of weld strengths shall be equal to or less than 0.20.

$$\text{Coefficient of variation} = \frac{\text{Highest weld strength - lowest weld strength}}{\text{Average weld strength}}$$

3.7.4 Weld strength test specimens.

3.7.4.1 Weld strength. Specimens tested shall exhibit strength equal to not less than 50 percent of the minimum parent metal strength.

3.7.4.1.1 Values. Weld strength values shall have a coefficient of variation not greater than the following:

Group A: 0.20
Group B: 0.25 in 90 percent of the specimens tested; 0.35 in the remainder
Group C: 0.35 in 90 percent of the specimens tested; 0.45 in the remainder

When combinations of different groups are welded, the less stringent values shall apply.

3.8 Welding schedule. The contractor shall establish suitable welding schedules for each material or combination of different materials and each thickness combination to be welded in production by an approved machine.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirement: of sections 3 and 5. The inspection set forth in this

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specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirement specified herein is classified as follows:

a. Qualification inspection (see 4.3) .

b. Quality conformance inspection (see 4.4) .

4.3 Qualification inspection.

4.3.1 Qualification of welding machines.

4.3.1.1 Qualification tests. Qualification tests of welding machines shall be performed to determine the consistency of operation of the machine. Each machine shall be qualified in accordance with 4.3.3 through 4.3.6. Qualification approval of the welding equipment and welds authorized herein shall be based on the consistent properties of the welds as shown by the contractor's tests.

4.3.2 Requalification. Requalification shall be required if the machine is rebuilt or relocated or if significant operational changes are made therein. Requalification shall be in accordance with 4.3.3 through 4.3.6.

4.3.3 Test material. Welding qualification shall be performed using 0.020 inch diameter nickel wire in accordance with type N-1 of MIL-STD-1276.

4.3.4 Test specimens. One set of test specimens shall be welded, comprising not less than 50 consecutive welds. Each specimen shall be welded at a 90° angle, ± 10° (see figure 1). No maintenance attention or control deviation shall be permitted during the welding of specimen sets except dressing of the electrode.

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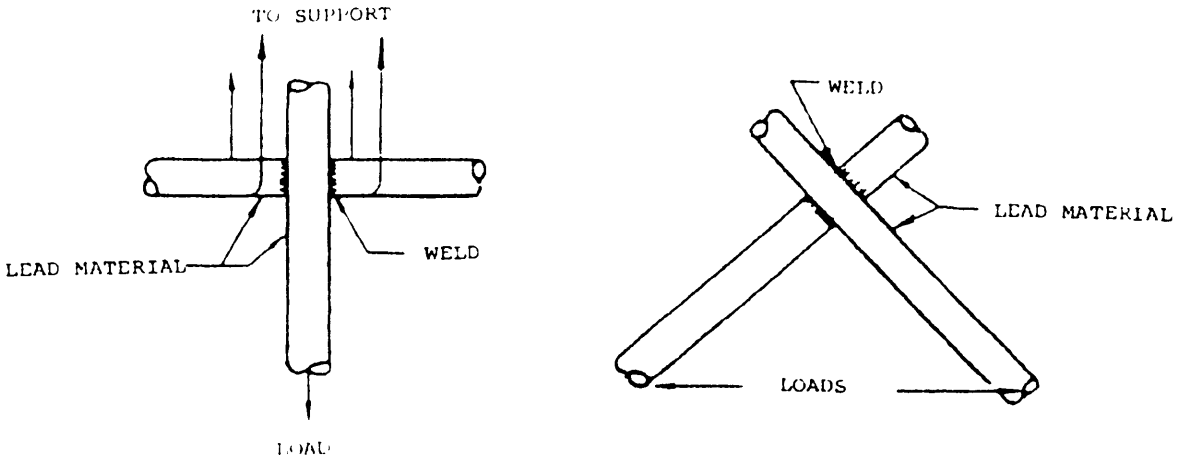



FIGURE 1. Test specimen welds.

4.3.5 Test procedure.

4.3.5.1 External appearance. All welds shall be examined for conformance with 3.7.1,1.

4.3.5.2 Metallographic. A minimum of two specimens from the set shall be subjected to metallographic examination. They shall be sectioned as nearly as possible through the center of each weld as shown in figure 2. The specimens shall then be polished and etched for metallographic examination and shall conform to 3.7.2. The specimens shall be prepared for metallographic inspection in accordance with ASTM E 3 and shall conform to 3.7.2.

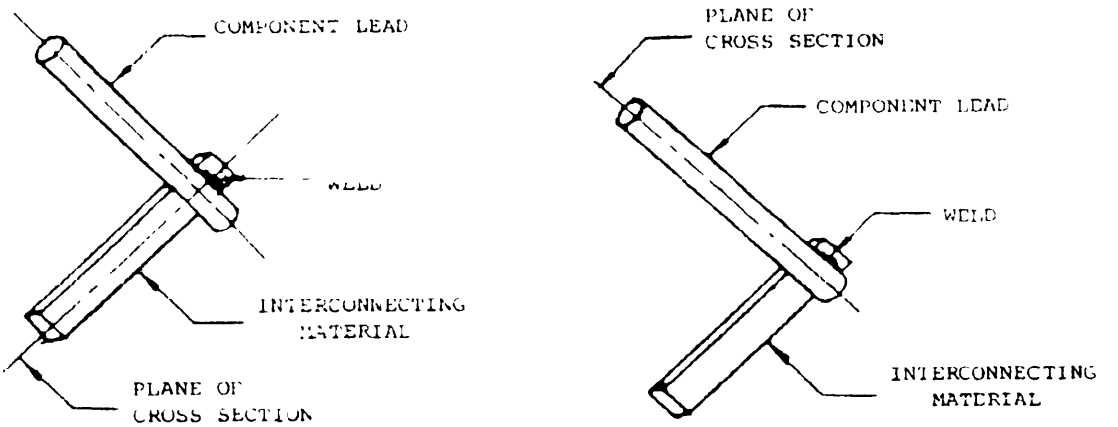



FIGURE 2. Test specimen welds for sectioning.

4.3.5.3 Weld strength. Forty-five specimens shall be tensile tested by either method shown on figure 1. However, the method selected shall be the same for all subsequent testing. Minimum parent metal breaking strength shall be determined by pull testing 20 specimens and selecting the lowest value of these. The strength of each weld shall conform to the requirements of 3.7.3.

4.3.5.3.1 Rate of loading. The specimen shall be loaded at a uniform rate, to be established by the contactor and used for all testing such rate shall be not greater than 60 inches per minute.

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4.3.6 Qualification records. The welding data shall be recorded on appropriate forms. These forms, approved by the authorized inspector, shall be readily available near the machines.

4.3.7 Certification of welding schedule.

4.3.7.1 Procedures. The welding schedule (see 3.8) shall be established prior to the welding of production parts and shall include the procedure for preparation of the material (including cleaning), all details of machine setup, and the control settings for the machine under consideration. The suitability of the welding schedule shall be established by the making and testing of not less than 25 welds. The welding schedules and test results shall be available to the Government for examination and evaluation.

4.3.7.2 Report forms. Appropriate report forms, if specified in the contract or purchase order (see 6.2), shall be prepared by the contractor to record the following information:

a. Welding machine make, model, and serial number
b. Size and composition of the material welded
c. Welding force
d. Energy setting
e. Polarities, types, and shape of electrodes and electrode holders
f. Equipment setup configuration
g. Results of tests.

4.3.7.3 Test procedure.

4.3.7.3.1 External appearance. All welds shall be examined for conformance with 3.7.1.1.

4.3.7.3.2 Metallographic. At least two specimens shall be examined metallographically through the center of the joint, one each in mutually perpendicular planes, as shown on figure 2. Both specimens shall conform to 3.7.2.

4.3.7.3.3 Weld strength. The remaining specimens, when tested in accordance with 4.3.5.3 and 4.3.5.3.1, shall meet the strength requirements specified in 3.7.4.

4.3.7.4 Process control data.

4.3.7.4.1 Similarity. Weld schedules developed on a qualified and approved machine may be used for each similar type of equipment, as long as process control data indicates that welds produced by the machines are within control limits.

4.3.7.4.2 Similar types of equipment. Similar types of equipment are those machines which are identical in the following respects:

a. Manufacture of machine
b. Type of machine, or model number
c. Electrical rating or capacity
d. Type of electrical energy

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e. Type of pressure application
f. Power cables, cable routing, and connections between power supply and welding lead.

4.3.8 Limits of weld strength. Limits of weld strength shall be established for each certified weld schedule, as specified in 3.7.4.1 and 3.7.4.1.1. Records shall be maintained and available for examination by the Government.

4.4 Quality conformance inspection. Quality conformance inspections shall be as specified in table I.

TABLE I. Quality conformance inspections.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Equipment | 3.4, 3.5 | 4.3.1, 4.4.2 |
| Weld standards | 3.7 | 4.3.7.2, 4.4.1 |
| External appearance of welds | 3.7.1 | 4.3.5.1 |
| Metallographic | 3.7.2 | 4.3.5.2 |
| Weld strength | 3.7.3.1 | 4.3.5.3 |

4.4.1 Production standards. Production welding shall be so accomplished as to obtain welds of uniform strength and of acceptable metallographic structure, rather than welds of maximum strength irrespective of metallographic structures. All test specimens used for checking production welding shall be representative of manufacturing practice. Specimens shall be tested by the contractor under the general supervision of authorized personnel. Prior to welding, all production parts shall be checked for conformance to 4.4.1.1 and 4.4.1.2.

4.4.1.1 Cleanliness. Areas to be welded shall be free from wax, dirt, oil, grease, or other foreign material detrimental to welding. Precautions shall be taken to avoid excessive handling of electronic pares and part leads.

4.4.1.2 Identification methods. Methods of identifying electronic parts or the polarity of part leads shall not interface with the welding process or the weldability of materials.

4.4.2 Equipment and operators. Production welding shall be performed on qualified welding equipment (see 4.3.1) in accordance with certified weld schedules (see 4.3.7). Prior to the operator's assignment to production welding, he shall have demonstrated, to the satisfaction of the contractor, a working knowledge of welding machines, weld schedules, drawings as related to the placement of components, interconnecting leads and location of welds, and a knowledge of component type and terminals. Test specimens prepared by the operator shall conform to 4.3.4.

4.4.3 Schedules. The contractor shall be responsible for the control of matching settings and all welding schedules. Records of all current schedules shall be available for examination by the authorized inspector.

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4.4.4 Sampling size and testing frequency. For production welds, metallographic examination of one sectioned sample per machine for each weld schedule per day is required. Production welding shall be limited to the 24 hour period following welding of a satisfactory metallographic specimen. For production samples that fail weld strength tests during a 24 hour period, testing of one newly welded metallographic section and three new strength test samples shall be accomplished successfully prior to resumption of welding. Each specimen shall conform to 3.7.1 and the control limits specified in 3.7.3.

4.4.5 Inspection of production parts .

4.4.5.1 External defects. All external welds shall be inspected for conformance with 3.7.1.

4.4.5.2 Functional testing. All welded electronic assemblies shall be functionally tested for conformance with applicable drawings or specifications after welding.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is concerned primarily with the processing of electronic modules used in missiles and aircraft. However, it may have application in the fabrication of electronic modules for general use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. When reports are required (see 4.3.7.2).

6.3 Electrode maintenance.

6.3.1 Electrode cleaning. Electrodes should be cleaned whenever splitting, metal sticking, deterioration, oxidation, or ether damage occurs.

6.3.2 Cleaning equipment. Electrodes may be cleaned with No. 400 emery paper or aluminum-oxide paper wrapped around the tip of a flat paddle. A circular motion should be used with the paddle inserted between the electrodes under slight electrode pressure.

6.3.3 Cleaning frequencies. Cleaning of tapered electrodes should be held to a minimum, since an increase in the surface area of the electrode tip



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will result. This change in surface area will produce changes in weld characteristics.

6.3.4 Replacement. Replacement of offset electrodes should be accomplished using a standard alignment tool to reposition the electrodes in the welding machine.

6.4 Weld resistance.

6.4.1 Resistance weld quality. The quality of a resistance weld, like any type of quality weld, is the result of the investigation and selection of various factors during the welding operation. These factors (or variables) in resistance welding are:

a. Composition and configuration of materials to be welded
b. Surface condition of the materials
c. Weld energy settings
d. Electrode pressure settings
e. Weld time
f. Type and configuration of electrodes.

6.5 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.6 Subject term (keyword) listing.

Adhesion
Attaching metals
Fusion
Soldering, machine
Weldments

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity
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Project 3439-A736



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-W-8939A(MI) | 20 July 1990 |

3. DOCUMENT TITLE
Welding, Resistance, Electronic Circuit Modules

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| U.S. Army Missile Command | (1) Commercial<br>(205) 876-6980 | (2) AUTOVON<br>746-6980 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Missile Command,<br>ATTN: AMSMI-RD-SF-TD-ST,<br>Redstone Arsenal, AL 35898-5270 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289 2340 | |

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